Re-Implementation of “LoRA: Low Rank Adaptation of Large Language Models”

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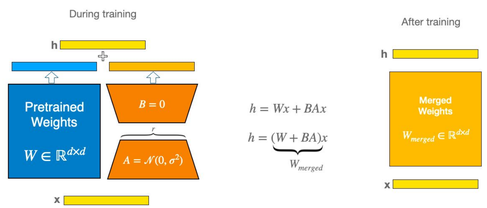
Introduction

Full Fine-Tuning (FFT) of Large Language Models (LLMs) demands significant VRAM and extended training times.



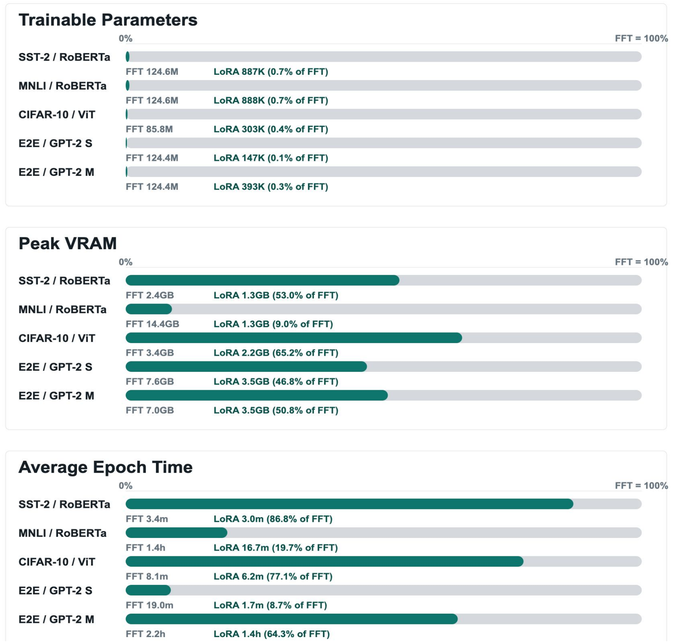
We aim to reduce trainable parameters and peak VRAM usage on a variety of transformer models while maintaining FFT accuracy.

Methodology

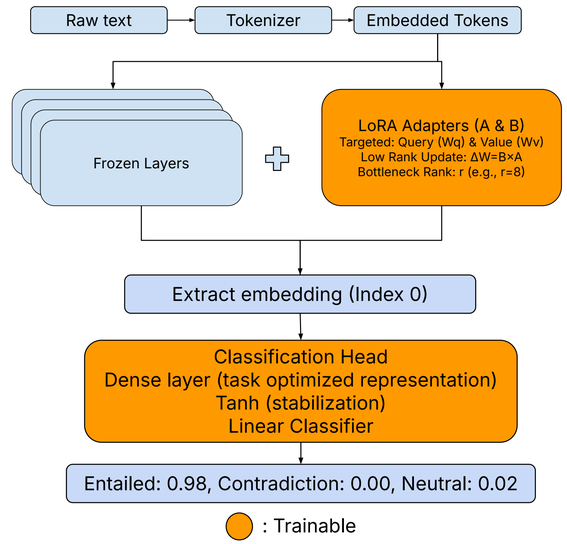
	RoBERTa	GPT-2	ViT
Hardware	A100	T4	T4
Experiment Type	Full FT vs LoRA	Small vs Medium	Generalization Test
Benchmark Dataset	SST-2 / MNLI	E2E NLG	CIFAR-10

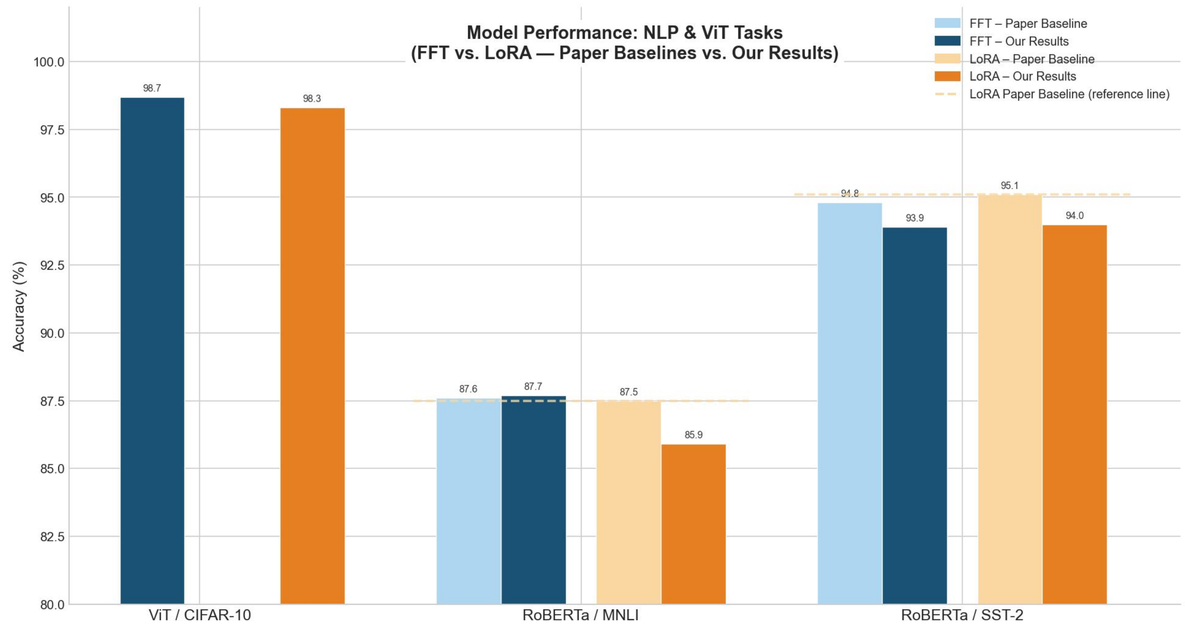
Modifications:

- Focus on smaller models
- Adjusting hyperparameters for speed
- Extended experiments to explore LoRA’s generalizability to images



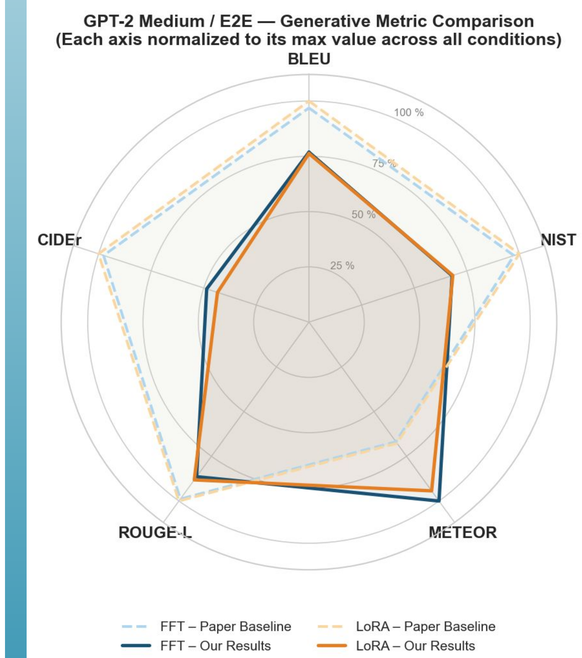
Premise: "The cat sat on the mat."
Hypothesis: "The animal is resting."





Conclusion

- Reduced trainable parameters by ~99%.
- Preserved accuracy on classification tasks
- Reduced peak VRAM and training time
- Possible sensitivity to task complexity, model scale, and hyperparameter tuning.
- LoRA is a practical, parameter efficient fine-tuning method when compute or memory is limited



Note: The METEOR metric is likely reporting higher scores than reality because our implementation reports the score compared to only the first reference.

References

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(4) Alexey Dosovitskiy et al. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.” In: International Conference on Learning Representations (CLR), 2021. arXiv:2010.11929. URL: https://arxiv.org/abs/2010.11929.
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